Multimedia Appendix

Samuel T Harrold, Yuimedi, Inc.

July 2026

Multimedia Appendix 2

Three-Pillar Organizational Assessment Rubric

This appendix presents the full Three-Pillar Assessment Rubric proposed in the main paper (section "The Three-Pillar Rubric"). Each indicator is scored as Low, Medium, or High Strength using evidence-based anchors from the literature reviewed in the main paper's Evidence Base section. Organizations scoring predominantly "Low Strength" across multiple pillars face the self-reinforcing degradation cycle that the framework identifies as the central threat.

Pillar 1: Analytics Maturity

Indicator	Low Strength	Med. Strength	High Strength	Evidence
HIMSS AMAM Stage	Stages 0-2: Fragmented data, limited reporting	Stages 3-4: Integrated warehouse, standardized definitions	Stages 5-7: Predictive analytics, AI integration	[1,2]
Self-service analytics	None; all analytics require IT intervention	Partial; BI tools available but underutilized	Widespread; clinical staff access data directly	[3,4]
AI/NL interface	No NL2SQL or conversational analytics	Pilot programs or evaluation underway	Natural language query capability deployed	[5,6]

Pillar 2: Workforce Agility

Indicator	Low Strength	Med. Strength	High Strength	Evidence
First-year analytics staff turnover	>30% (High instability)	15-30%	<15% (High stability)	[7,8]
Avg. leadership tenure	< 3 years	3-5 years	> 5 years	[9]
Knowledge concentration	Critical expertise held by 3 or fewer individuals	Partial documentation; some cross-training	Distributed expertise; documented processes	[10,11]

Pillar 3: Technical Enablement

Indicator	Low Strength	Med. Strength	High Strength	Evidence
Data access	SQL/technical expertise required for all queries	IT queue for complex queries; basic self-service	Natural language or visual query interfaces	[4,6]
Interoperability	Fragmented systems; manual reconciliation	Partial integration; some automated feeds	Unified data platform; real-time integration	[12,13]
Schema coupling	Hard-coded reports break on schema change	Partial semantic layer; some automated feeds	Semantic layer adapts; CI/CD detects drift	[14,15]

References

1. HIMSS Analytics. *Analytics maturity assessment model (AMAM) global report. Healthcare Information and Management Systems Society*. HIMSS Analytics; 2024. https://www.himss.org/maturity-models/amam/

2. Healthcare IT News. HIMSS launches modernised Analytics Maturity Assessment Model. Published online 2024. https://www.healthcareitnews.com/news/asia/himss24-apac-adoption-model-analytics-maturity-gets-facelift

3. Health Catalyst. The healthcare analytics adoption model: A roadmap to analytic maturity. Published online 2020. https://www.healthcatalyst.com/learn/insights/healthcare-analytics-adoption-model-roadmap-analytic-maturity

4. Shahbaz G M. Investigating the adoption of big data analytics in healthcare: The moderating role of resistance to change. *Journal of Big Data*. 2019;6(1). doi:10.1186/s40537-019-0170-y

5. Ziletti &D A. Retrieval augmented text-to-SQL generation for epidemiological question answering using electronic health records. *NAACL 2024 Clinical NLP Workshop*. Published online 2024. https://arxiv.org/abs/2403.09226

6. Yuan C, Ryan PB, Ta C, et al. Criteria2Query: a natural language interface to clinical databases for cohort definition. *Journal of the American Medical Informatics Association*. 2019;26(4):294-305. doi:10.1093/jamia/ocy178

7. NSI Nursing Solutions. *2025 National Health Care Retention & RN Staffing Report*. NSI Nursing Solutions; 2024. https://www.nsinursingsolutions.com/documents/library/nsi_national_health_care_retention_report.pdf

8. Hackney A. *Onboarding New Hires in the Clinical Informatics and Practice Support (CIPS) Department With the Tiered Skills Acquisition Model (TSAM): A Program Evaluation*. PhD thesis. ProQuest Dissertations; 2024. https://search.proquest.com/openview/9a3a9ef301c01ff0129d07f685b8643f/1

9. WittKieffer. *CIO Insights: The State of Healthcare IT Leadership*. WittKieffer; 2024. https://api.wittkieffer.com/wp-content/uploads/2012/10/cio-insights-the-state-of-healthcare-it-leadership-wittkieffer-october-2024.pdf

10. Massingham PR. Measuring the impact of knowledge loss: A longitudinal study. *Journal of Knowledge Management*. 2018;22(4):721-758. doi:10.1108/JKM-08-2016-0338

11. Foss NJ. The emerging knowledge governance approach: Challenges and characteristics. *Organization*. 2007;14(1):29-52. doi:10.1177/1350508407071859

12. Gal MS, Rubinfeld DL. Data Standardization. *NYU Law Review*. 2019;94(4):737-770. https://www.nyulawreview.org/issues/volume-94-number-4/data-standardization/

13. Zhang Y, Callaghan-Koru JA, Koru G. The challenges and opportunities of continuous data quality improvement for healthcare administration data. *JAMIA Open*. 2024;7(3):ooae058. doi:10.1093/jamiaopen/ooae058

14. Mannapur SB. Understanding data drift and concept drift in machine learning systems. *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*. 2025;11(1):318-330. doi:10.32628/CSEIT25111239

15. Battula SKY. Adaptive Data Quality Management for Multi-Cloud Healthcare Warehouses: FHIR-Aware Semantics and Unsupervised Thresholding. *International Journal of Artificial Intelligence, Data Science, and Machine Learning*. 2025;6(4):218-226. doi:10.63282/3050-9262.IJAIDSML-V6I4P130